Know When to Fold: Risk-Controlled Abstention for Protein–Ligand Co-Folding under Novelty Shift

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Abstract

A practitioner who acts on a predicted protein–ligand pose needs to know when to trust it. Co-folding models (AlphaFold3, Boltz, Chai) report confidence scores that correlate with pose accuracy, but a correlation is not a decision rule, and it is weakest on exactly the novel pockets and chemotypes a discovery program pursues. We build `foldgate`, a training-free layer that turns frozen co-folding confidence into an accept/abstain gate with a finite-sample error guarantee, and we ask where that guarantee can and cannot be kept. Calibrating on familiar targets and deploying on novel ones drives the realized error among accepted poses to 0.55 (AlphaFold3, 90% CI $[0.52, 0.58]$; 0.48 to 0.66 across five models) against a 0.20 target. A per-stratum feasibility analysis shows this is not a tuning problem, and not an artifact of a weak score: across a frozen score family — the global ranking score, interface ipTM, and the ligand-local pLDDT that is the field's actual pose-triage signal — 13 of the 40 non-reference model-by-novelty-stratum cells admit *no* threshold on *any* of them that reaches the target at any coverage at $\alpha = 0.20$, so on structurally novel targets abstention is the only certified action. An exact decomposition of the error explains why — the part that comes from confidence meaning something different on novel molecules is invisible to any label-free reweighting — and turns into a diagnostic that picks the repair from a small labeled probe. Group-conditional recalibration restores per-stratum control (AlphaFold3 certifies 52% and 39% of the moderately-novel strata, against 8% and 12% before), and marginally, under leakage-free target-grouped cross-validation, its combined-feature gate retains 73% of AlphaFold3 poses with the certificate holding in four of five held-out folds where the raw-confidence gate certifies almost none. We price the repair honestly: extending certification one novelty stratum outward costs on the order of 80 in-stratum crystal structures, the raw-confidence gate is CPU-only but the combined-feature gate consumes five diffusion samples and cross-model runs, and the most novel targets admit no repair by stratification at all. The layer runs on released predictions and ships a certificate card per model and novelty stratum that reads FEASIBLE, ABSTAIN-and-abandon, or ABSTAIN-and-collect-labels.

1 Introduction

Co-folding models predict a protein–ligand complex and emit confidence scores (interface predicted TM-score, ipTM; ranking score; predicted aligned error). A practitioner who wants to act on a predicted pose needs a decision rule that carries an error guarantee. Conformal selective prediction supplies one: accept a pose when its score clears a threshold, abstain otherwise, and certify that the error rate among accepted poses stays below a target α with high probability [7, 23]. The catch is that this guarantee assumes exchangeability, that a new target is statistically interchangeable with the calibration examples, and that breaks precisely where deployment matters, on ligands and pockets dissimilar to the training set. In drug-discovery terms the model is most overconfident on the never-before-seen molecules a program cares about, so the operational question is not average accuracy but whether a single delivered pose can be trusted.

We make three contributions, all training-free with respect to the co-folding model and all computed from released predictions on CPU. **(1)** We give a finite-sample selective-risk gate over frozen co-folding confidence, show its guarantee collapses under novelty shift, and map the collapse with a per-stratum *feasibility frontier* that decays to zero on novel strata; reliability drift attributes the collapse to concept shift (Sec. 3–4). **(2)** We prove an exact decomposition of the realized selective risk into a covariate-corrected reference and an accept-region concept gap that no training-free reweighting can move, turning it into a measurable diagnostic that selects, from a labeled probe, which repair a deployment region admits (Sec. 5). **(3)** We repair what is repairable with group-conditional calibration and a label-free worst-subpopulation certificate, match the certifier to the loss, and *price* the repair: the label cost as a design curve, a deployment-computable proxy stratifier, and the honest bottom line that the extreme novel tail is uncertifiable by stratification alone (Sec. 6). A leakage-free utility evaluation and a decision-curve analysis follow (Sec. 7).

The closest prior work is CoDrug [14], which conformalizes a *scalar* molecular property under covariate shift with the same weighted-conformal tool [3] our decomposition shows concept shift defeats in this domain; we are not aware of a prior conformal treatment of co-folding pose correctness, or of training-set similarity used as the shift variable for it. Confidence-tracks-accuracy studies for co-folding virtual screening [10] rank by ipTM but supply neither a coverage guarantee nor an abstention rule, which is the gap this layer fills. A reweighting-invariant selective-risk floor of the same abstract character was shown independently and concurrently for LLM abstention [19] (arXiv preprint, June 2026); ours is its co-folding specialization, localized per novelty stratum with a matching in-stratum achievability bracket. Conformal selection [15, 16] (the latter an arXiv preprint) is complementary: it certifies which *compounds* to advance given a labeled calibration set, while this layer certifies the *pose* correctness a co-folding screen rests on, training-free.

We connect four literatures. Using training-set similarity as a reliability predictor is the cheminformatics *applicability domain* [40–42], and our claim is narrowly the conformal treatment of co-folding *pose* correctness with that similarity as the shift variable, not the idea of similarity-as-reliability. The abstention gate is selective prediction [7, 43] and its learning-to-defer descendants [44, 45]. The failure of exchangeability under shift is exactly the regime of conformal prediction beyond exchangeability [33], which bounds the coverage gap by a total-variation distance and is the alternative route to our problem; adaptive conformal [34] and label-shift conformal [35] address the online and label-shift variants we do not. On the repair side, multicalibration and multivalid prediction [36, 37] handle overlapping, computationally-identifiable groups rather than a fixed partition, a strictly more flexible alternative to the Mondrian repair we deploy, and the DRO certificate follows Duchi and Namkoong [38] and Cauchois et al. [24]. Net-benefit analysis is due to Vickers and Elkin [39].

2 Data and definitions

Unit and loss. The decision-relevant unit is a delivered top-1 pose per system with a native confidence s. The loss is $L = \mathbf{1}[\text{symmetry-corrected ligand-RMSD} > 2\,\text{Å}]$ against the released BiSyRMSD label (binding-site-superposed, symmetry-corrected ligand RMSD: superpose on the pocket, then measure ligand-atom RMSD up to molecular symmetry), so a certified selective risk is a bound on the accepted-set error rate. We use L throughout (Table 1); recomputed geometry is used only where noted (Table 2).

Novelty axes. We stratify by training-set similarity along two structural axes: a ligand axis (ECFP4 Morgan Tanimoto to the nearest training ligand; a Tanimoto near 1 means an almost-identical training ligand exists) and a pocket axis (SuCOS shape $\times$ pocket query-coverage, a shape-overlap score weighted by how much of the query pocket is covered), both released by Runs N' Poses as similarity to the public pre-cutoff PDB. Each axis is binned into quartiles S_0 (familiar) to S_3, with systems for which the released similarity search returns no neighbor (missing similarity, about 3% of poses, AlphaFold3 $n \approx 76$) forming a fifth *no-computable-analog* stratum S_4. We are careful with S_4: on inspection it is a mixture, because the similarity is missing for the whole system rather than the ligand alone, and roughly half of S_4 is multi-ligand-chain systems or ubiquitous cofactors and sugars (ADP, FAD, NAG, …) that certainly have training analogs but fall outside the shipped search, so their missing similarity is a coverage gap, not novelty. The genuinely drug-like, single-component, non-cofactor core is ~ 29 of the 76 instances (its pose accuracy 0.44 against 0.67 elsewhere confirms it is genuinely hard). We therefore treat S_4 as a diagnostic extreme rather than a headline, report its composition, and note that externally recomputed ECFP4/pocket similarity would purify it to the ~ 29-member core; the load-bearing break is carried by S_3, which has no such ambiguity. The quartile edges are taken from the full-benchmark *similarity* marginal (a covariate, not a label, so this is not label leakage), and the frontier is robust to replacing them with fixed absolute similarity cuts, which we promote as an alternative (the zero-frontier fraction stays near 0.4 across $3/5/7$ bins and fixed edges); Table S1 lists n and the similarity range each bin spans, per model and axis. One similarity reference stratifies all five governed models because they share the same ~ 2021 training cutoff, so a ligand or pocket novel to one is novel to the others; the

Table 1: Notation.

L	loss $\mathbf{1}[\mathrm{RMSD} > 2\,\text{Å}]$	α, δ	risk target, failure prob. $(0.20, 0.10)$
s, τ	score, accept threshold ($s \geq \tau$)	c	coverage (accepted fraction)
ν, S_ν	novelty axis, stratum	η_P, η_Q	source/target score-conditional error $P(L{=}1 \mid s)$
R_{ref}	covariate-corrected reference risk	$\bar{\Delta}_c$	accept-region score-fiber gap
$D(\nu)$	reliability drift at stratum ν	$m^\star, \beta^\star$	certified min. subpop. mass, CVaR level
$\rho^\star$	DRO robustness radius	n_g	in-stratum labels

Table 2: Label provenance. The primary selective-risk results use the shipped BiSyRMSD label. The protomer trap of App. C afflicts only cross-frame *superposition*, so the cross-model danger floor is restricted to the single-chain subset; interaction-fingerprint recovery enumerates contacts *within* each structure and matches them by residue identity (seqid, resname), which is invariant to a protomer swap (homodimer copies share numbering), and its result is unchanged on the single-chain subset (gate coefficient $+0.04$–$+0.06$ there vs. $+0.03$–$+0.07$ on the full set).

Result	Label / geometry	n
Break, frontier, repair, utility (Secs. 3–7)	shipped BiSyRMSD	11,254 target-labels
Cross-model danger floor / consensus (SI)	recomputed, single-chain receptors	6,163 instances
Interaction-fingerprint recovery (Sec. 7)	per-structure contacts vs. crystal	12,475 poses

reference year is not incidental (re-keying the pocket similarity to a 2023 PDB snapshot rank-correlates with the 2021 one at only Spearman 0.14, since the PDB grows), which is precisely why we match the reference to the cohort's cutoff and analyze the single 2023-cutoff model (Boltz-2) separately with its own 2023 reference. We treat structural similarity, not deposition recency, as the operative shift variable, for a reason the data forces (Sec. 3).

Cohort. We evaluate five models (AF3 [11], Boltz-1 [20], Boltz-1x [21], Chai-1 [12], Protenix [13]) on Runs N' Poses [8]. Figure 1 is the reduction chain: 13,535 raw delivered poses (six models) minus the 933 Boltz-2 [22] poses used only as an ungoverned affinity comparator (Sec. 7) give 12,602 governed poses over 2,425 systems; deduplicating to one pose per (system, model) gives 11,254 independent target-labels for the five governed models. Certificates that must be per deployed model use the target-label unit, since several ligand instances of one system fail together and pooling them as independent draws is anti-conservative. The cohort is chemically diverse and not a single-family monoculture: 55% drug-like (300–500 Da) with a 24% fragment tail, 954 receptor sequence clusters (the largest covering 7% of systems), and no PDB header class above 17%.

Deployment cost. The native-score gate is CPU-only and consumes one released prediction per target. An optional combined score (a calibration-only gradient-boosted map from cheap per-prediction features to $P(\text{correct})$: confidence, interface ipTM, ensemble spread, cross-model agreement, PoseBusters validity, ligand descriptors) improves the operating point but consumes the five-sample ensemble and cross-model features, so it needs N model inferences per target. We report the native-only guarantee prominently and flag the combined score as the upgrade. The combined score's cross-model features are moderately correlated with novelty ($|\rho|$ up to 0.35), so it is a partly novelty-aware re-score; this is legitimate and places the *combined* score on the achievability side of the theorem below, which governs the frozen *native* score.

A benchmark-integrity finding, of independent value. Anyone recomputing pose geometry on Runs N' Poses will hit a silent trap we document in full in App. C: the benchmark ships only the receptor chain while co-folding models predict the full assembly, so a chain-ordinal superposition can seat the ligand in the wrong protomer of a homodimer, and a distance-consistency check passes at 0 of 13,146 pairs while 15% of the recomputed binary labels are wrong; only comparison against the shipped BiSyRMSD exposes it, and restricting to single-chain receptors raises agreement from Spearman 0.62 to 0.995. We surface it here rather than bury it because it is a reusable finding for the benchmark, and we report which of our own results touch recomputed geometry (Table 2).

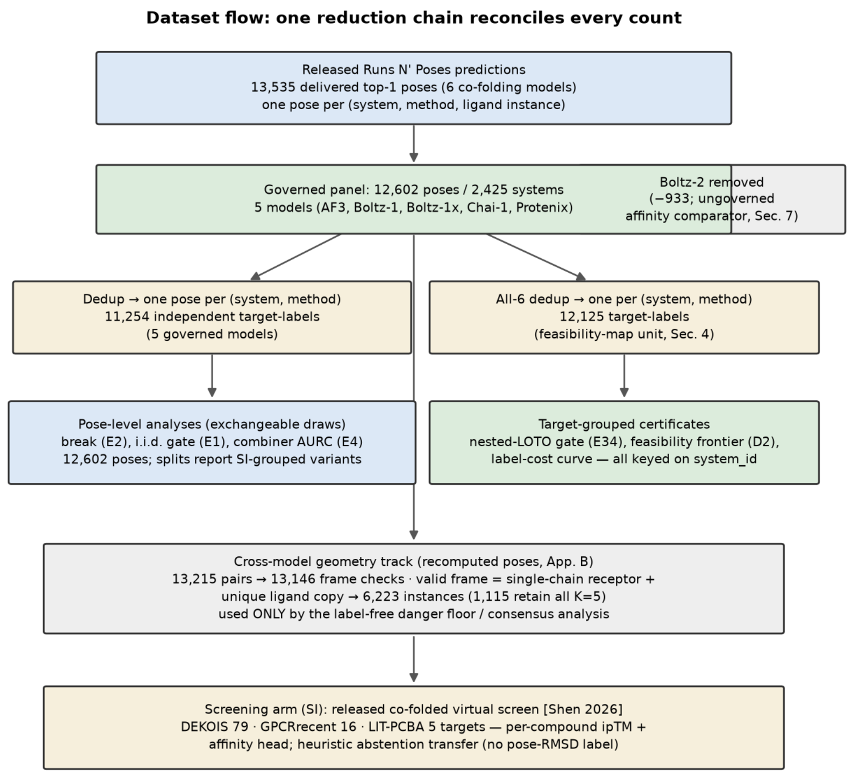

Figure 1: Dataset flow. One reduction chain reconciles every count; each node is annotated with the experiments that consume it. Counts are recomputed live from the released predictions.

Gate. The gate accepts when $s \geq \tau$ and picks the largest accept set whose selective risk is certified $\leq \alpha$ by Learn-then-Test with fixed-sequence testing [1], testing each $H_0(\tau) : P(L{=}1 \mid s \geq \tau) \geq \alpha$ with an exact binomial tail along a pre-specified, data-independent coverage grid (top-k ranks, $c \in \{0.05, 0.06, \ldots, 1.00\}$, fixed before any risk was computed and archived in the repository). This gives $P(\text{selective risk} \leq \alpha) \geq 1-\delta$, finite-sample and distribution-free.

3 The exchangeability break

The break. A single global threshold hides severe per-stratum under-coverage. For AF3 the marginal realized risk $R_{\text{marg}} = 0.177$ masks $R_{\text{marg},S_3} = 0.375$ on the novel ligand stratum (Fig. 2, with Clopper-Pearson intervals and accepted n per point). The operational failure is starker: calibrating the gate on the familiar strata (S_0, S_1) and deploying on the novel tail (S_3, S_4) drives the realized error to $R_{\text{transfer}} = 0.55$ (AF3, 90% CI $[0.52, 0.58]$, $n = 734$; 0.48–0.66 across the five models) against the 0.20 target, so the certificate is inverted, not merely loose. We lead the break on S_3; the no-computable-analog S_4 (AF3 $n \approx 76$, greyed in the figure) is small-sample and enters only in the feasibility frontier, where zero coverage is the point.

The break is a base-rate collapse, not a signal collapse. A natural worry is that the score simply carries no information on novel targets. It does not: within S_3 the native score still ranks correct from incorrect poses well above chance — AUROC 0.71 (AF3, 90% CI $[0.68, 0.75]$), 0.64 (Protenix), 0.56 (Chai) — so the gate is not anti-selective. What breaks is the base rate: correctness on S_3 falls to 0.44 (AF3), so even a well-ranking score cannot place a threshold whose accepted set clears α at usable coverage. The score's discriminative signal only collapses to chance on the extreme tail (AF3 S_4 AUROC

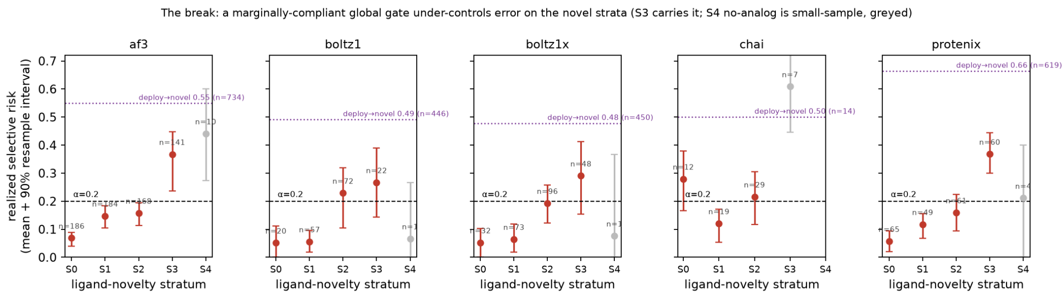

Figure 2: The break. Per-stratum realized selective risk of a marginally-compliant global gate, over 300 target-grouped calibrate/deploy resamples (mean, 90% resample interval, median accepted *n* per point); the no-analog S_4 is greyed as small-sample. Realized risk climbs above the target $\alpha = 0.20$ as ligand similarity falls, and the dotted line marks the calibrate-on-familiar / deploy-on-novel realized risk (AF3 0.55). The break is carried by S_3.

0.54, CI $[0.43, 0.66]$; Protenix S_4 0.50). This is the sharper statement of the break: the frozen score remains a usable *ranker* on novel strata but not a usable *certifier*, because certification is a statement about the accepted-set error rate and that is base-rate-bound.

Why it breaks: concept shift, not recency. Weighted conformal would fix this if the shift were pure covariate shift, where $P(L \mid s)$ is stable and only the marginal $P(s)$ moves. We test that by measuring *reliability drift* $D(v) = \sum_{\text{bins}} w\,[P(L{=}0 \mid s, S_0) - P(L{=}0 \mid s, S_v)]$, the confidence-conditional correctness gap between the familiar reference and stratum v. Drift is large on structural novelty and its 90% bootstrap interval excludes zero for every model (Protenix reaches $+0.63$ on pocket novelty, Chai $+0.52$); the $+0.63$ is not a sparse-bin artifact, as every confidence quintile behind it holds ≥ 103 novel-pocket targets and the top-confidence bin still recovers to only 0.54 correct against 0.95 on the familiar stratum. Confidence is optimistic on novel structure: at the same reported score, poses in the novel stratum are less often correct. A signed accept-region decomposition makes this quantitative: for AF3 on S_3 the target-risk gap of 0.328 splits into a score-fiber term of 0.309 (90% CI $[0.27, 0.35]$, resampling targets) and a coverage-alignment term of 0.018, which Sec. 5 turns into an identity. The effect is not an artifact of the differing per-model composite scores: recomputed on a common interface-ipTM score definable for all five models the S_3 drift is large and positive for every model (Protenix $+0.63$ pocket, $+0.50$ ligand; Chai $+0.56$ pocket; CIs far from zero), and the two most overconfident models (Protenix, Chai) rank the same on either score, so we report cross-model magnitudes but not fine-grained mid-pack ordering.

The temporal axis is a benchmark artifact, not a null result. Earlier framings contrasted a large structural drift with a near-zero temporal drift. That contrast is real but must be stated precisely: every Runs N' Poses structure is deposited after the 2021-era models' training cutoff, so the temporal axis ranks recency *among already-out-of-training structures* rather than in- versus out-of-training, and a flat drift there is a property of the benchmark, not evidence of temporal robustness. The one model with a genuine within-benchmark cutoff boundary is Boltz-2 (2023-06-30), which is outside the governed five-model cohort and enters only here: of its 933 poses, 77 predate the cutoff and 856 follow it, and the reliability drift across that boundary (on its *pose* ranking confidence, not the affinity head) is $+0.005$ (90% CI $[-0.075, 0.079]$), while its structural S_3 break is unchanged whether keyed on the 2021- or the correctly-referenced 2023-training similarity (0.364 vs. 0.367 correct). This is a single-model observation with 77 in-cutoff targets, not a cohort result; we report it as the one available genuine temporal test and do not claim a temporal-generalization result. Structural similarity is the operative shift variable throughout. A protein sequence-identity axis is likewise uninformative here, but for the same benchmark-redundancy reason: $\sim$37% of Runs N' Poses systems sit at the sequence-identity ceiling, so its novelty quartiles collapse and produce no break (worst-stratum excess -0.02 against $+0.24$ ligand, $+0.29$ pocket); this is a property of the benchmark's homology redundancy, not evidence that sequence novelty is safe, and a low-homology benchmark is needed to test it.

4 The feasibility frontier

The break is a fixed-coverage statement; it leaves open whether α is reachable by lowering coverage, so we measure that directly. We fix the deployed rule on the familiar stratum and, per target stratum, report the largest source coverage $c_g^\star = \max\{c : R_{Q,g}(\tau(c)) \le \alpha\}$ at which it still holds α, one delivered pose per (system, model). The frontier decays with novelty and then vanishes: for AF3 on the ligand axis $c^\star$ runs $1.00, 0.85, 0.75, 0.20, 0.00$ across S_0 to S_4 at $\alpha = 0.20$, and $0.65, 0.40, 0.10, 0.00, 0.00$ at $\alpha = 0.10$. The frontier is computed per governed model, one delivered pose per (system, model); the all-six dedup pool of Fig. 1 (12,125 target-labels) includes the ungoverned Boltz-2, which is excluded from these cells. Of the 50 model-by-stratum cells (two axes $\times$ five governed models $\times$ five strata), the ten S_0 reference cells are feasible by construction (the rule is calibrated on S_0: $c^\star = 1.00$ at $\alpha = 0.20$, 0.65 at $\alpha = 0.10$), leaving 40 non-reference cells. Requiring at least 20 accepted targets to call a stratum feasible, on the global ranking score **22** of those 40 admit no operating point at any coverage at $\alpha = 0.20$ (of which 14 survive a per-cell exact binomial lower bound on $R_{Q,g} > \alpha$ at every coverage, 13 a family-wise Holm or Benjamini-Yekutieli correction). But the count is *score-dependent*, because the frontier is bounded by the accepted-set base rate, not by ranking quality (Sec. 3): computed identically on interface ipTM it is 16, and on ligand-local pLDDT — the field's actual pose-triage signal, extracted from the predicted structures' per-atom confidence and a materially better ranker (AUROC 0.72–0.85 against 0.59–0.70 for the global score) — it is 18. We therefore state the impossibility across the frozen score family: **13** of the 40 cells admit no threshold on *any* of these scores (global ranking, interface ipTM, mean and worst-quartile ligand pLDDT) that reaches α at any coverage, and 21 of 40 at $\alpha = 0.10$. Ligand pLDDT is a better ranker yet does not close the frontier precisely because the break is base-rate-bound, not discrimination-bound; the one pose-triage signal we could not test, the protein-ligand PAE block, is absent from the released prediction dump (it ships coordinates and ranking scores only) and would require regeneration. The infeasibility is not an artifact of the quartile binning: the zero-frontier fraction stays near 0.4 whether we use 3, 5, or 7 strata or fixed-similarity edges. The deployed view is starkest of all: the AF3 rule that Learn-then-Test certifies at $\alpha = 0.20$ on familiar targets realizes risk $R_{\text{gate},S_3} = 0.538$ on S_3, a margin of -0.338.

A one-sided escape survives the impossibility, which constrains only certified *upper* bounds: cross-model pose disagreement supplies a label-free *lower* bound on ensemble risk by triangle inequality to the crystal pose, extending an oracle-free disagreement floor known for discrete outputs [27, 28] and the folklore that pose agreement filters wrong poses [29] to a continuous structured target. It is blind to consensus error, where all models share one wrong mode: the consensus regime covers 64% of the analyzed complexes at risk 0.152 against 0.698 on the diverse remainder, and the hidden error grows from 0.058 on S_0 to 0.120 on S_3 even as consensus mass shrinks. The construction and its single-chain provenance are in the SI.

5 An exact decomposition, and what a training-free reweighting cannot fix

The contribution of this section is a *diagnostic*: a single quantity, measurable from a small labeled probe, that tells a practitioner in advance whether a label-free repair (weighted conformal) will work or whether only in-stratum recalibration will. The supporting fact is an exact identity, a one-line tower-property lemma; we state it first and then show it is the diagnostic. Fix the frozen score s and the accept rule $\{s \ge \tau\}$; write $Y = L$ for the error label. We work in the score's own σ-algebra $\sigma(s)$, because that is what a gate on s and a covariate-measurable weight on s can see: let $\eta_P(s) = P_P(Y{=}1 \mid s)$ and $\eta_Q(s) = P_Q(Y{=}1 \mid s)$ be the score-conditional error rates under the calibration law P and the deployment law Q, and call a reweighting *training-free* if its weights are $\sigma(s)$-measurable and use no target labels (density-ratio, classifier, or KLIEP weights on s, including the point mass of weighted conformal). Write $R_{\text{ref}}(\tau_c) = \mathbb{E}_Q[\eta_P(s) \mid s \ge \tau_c]$ and $\bar{\Delta}_c = \mathbb{E}_Q[\eta_Q(s) - \eta_P(s) \mid s \ge \tau_c]$. We estimate $\bar{\Delta}_c$ by binning s (Sec. 3), so it is the accept-region gap between the source and target label rates *at the same score*. Whether that gap is a change in $P(Y \mid \text{full covariate})$ (concept shift proper) or covariate shift within a score fiber is not identifiable from s alone; either way it is invisible to any $\sigma(s)$-measurable reweighting, which is exactly what the impossibility requires, so we name it *score-conditional reliability drift* rather than concept shift, and report its binning sensitivity.

Lemma (exact decomposition; App. A). At fixed coverage c the realized target selective risk is $R_Q(\tau_c) = R_{\text{ref}}(\tau_c) + \bar{\Delta}_c$, and $\bar{\Delta}_c$ is invariant to the choice of training-free ($\sigma(s)$-measurable) weights. Hence the oracle-weighted plug-in certificate reports R_{ref} and under-states the realized risk by exactly $\bar{\Delta}_c$.

We deliberately do not dress this as an impossibility theorem: "α is achievable at coverage c iff $R_Q(\tau_c) \leq \alpha$" is a definitional rearrangement, and whether α is reachable by *lowering* coverage is the empirical feasibility frontier of Sec. 4, not a corollary. The content is that the accept-region score-fiber gap $\bar{\Delta}_c$ is *measurable and invariant*: the part of the error that comes from the score meaning something different on novel molecules is exactly the part no unlabeled reweighting can see or remove. Two distinct reasons are at work and we keep them apart: a $\sigma(s)$-measurable weight cannot change the within-fiber label rate (the tower-property identity below), while any weight at all, even a full-covariate one, that realizes coverage c accepts the identical set and so cannot move the realized risk (coverage pinning). The proof is a tower-property identity: a $\sigma(s)$-measurable weight recovers only the source score-conditional η_P and cannot reach η_Q, completing the covariate-shift boundary of weighted conformal [3] and analogous in spirit to the irreducible term of domain-adaptation lower bounds [25]; a reweighting-invariant floor of the same character was shown concurrently and independently for LLM abstention [19], and our contribution is the co-folding instantiation with per-stratum localization, not priority on the abstract floor. Matching achievability: in-stratum recalibration attains the floor R_{Q_g} given n_g stratum labels; pointwise conditional coverage being impossible [6], the guarantee is neighborhood-conditional, and its ratio functional is controlled with high probability by Learn-then-Test, whose finite-sample cost is exactly the label-cost curve of Sec. 6 rather than an asserted rate.

The decomposition's equalities assume the score has no atom at the operating threshold. On the primary `ranking_score` gate this holds exactly in the data (the ties mass at the deployed τ is 0 for every model); on the interface-ipTM feature the boundary-randomization bracket is at most 0.7% of coverage. So the identity is exact for the deployed native gate, with a $\leq 0.7\%$-coverage bracket noted where ipTM is used.

Validation. A synthetic generator with a known $P(Y \mid s, \nu)$ reproduces every term: $R_{\text{ref}} = 0.273$ and $\bar{\Delta} = 0.205$ sum to the realized $R_Q = 0.478$, the invariance of the realized risk across nine reweightings (residual < 0.001) is a regression test of the decomposition, and sweeping the target drift locates the coverage at which no reweighting reaches α. On the public non-co-folding benchmarks elec2 (worst-block concept gap 0.36) and ACS Income (gap 0.05), a single-threshold gate under-controls the worst block, weighted conformal issues a certificate its realized risk then violates on elec2 but repairs on ACS, and group-conditional calibration certifies control by abstaining, with the elec2/ACS realized worst-block risks in App. B.

Drift selects the repair. Reliability drift $D(\nu)$ is a label-based diagnostic that tracks $\bar{\Delta}_c$, so it decides, from a labeled probe rather than after a failed deployment, which repair a region admits. A small drift signals a covariate-dominated shift that weighted conformal repairs by choosing a more conservative threshold from unlabeled data (weighted conformal's *only* lever is threshold selection; on the moderate ligand shift $S_0, S_1 \to S_2$ it pulls AF3 realized error from 0.27 at 98% coverage to 0.19 at 70% coverage). A large drift signals a concept-dominated shift where weighted conformal silently violates its own certificate, leaving group-conditional recalibration as the only valid repair.

6 Repairing the guarantee, and pricing it

Group-conditional calibration [17] keys a separate threshold per novelty stratum, restoring per-stratum control at an honest coverage cost (with the combined score AF3 accepts 52% of S_1 and 39% of S_2 at risk $\leq \alpha$, against 8% and 12% natively; Fig. 3). Where the marginal gate over-shoots α on the novel strata the per-stratum gate holds risk at or below it and, on the no-analog tail where no certified slice remains, folds to zero coverage rather than issue a false certificate. The repair does not require the gradient-boosted combiner: a per-stratum Venn–Abers recalibration of the native score reaches the same stratum-conditional guarantee (AF3 realized risk 0.166 at 24% coverage vs. 0.190 at 18% for the group-conditional combiner), while the field baselines break under the same shift, a fixed ipTM≥ 0.8 gate at

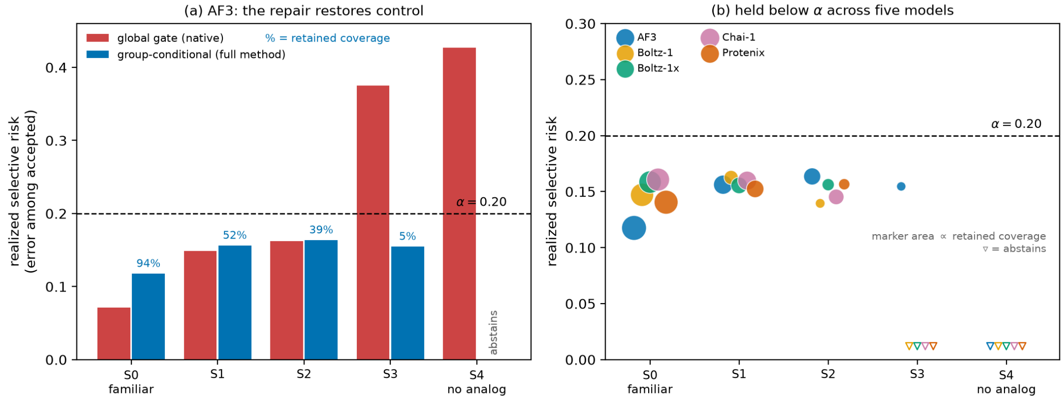

Figure 3: The repair. Group-conditional calibration holds realized selective risk at or below $\alpha = 0.20$ on the strata where the marginal gate over-shoots, folding to zero coverage on the no-analog tail rather than issuing a false certificate. Coverage labels are direct; the coverage cost grows with novelty.

Table 3: Baselines under the novelty shift (AF3, calibrate on familiar, deploy on novel; realized risk / coverage at $\alpha = 0.20$). The field baselines break; every gate that keys calibration on novelty holds by abstaining. RLCP reaches lower risk at higher coverage, but its guarantee is randomized and marginal-over-localization, not the stratum-conditional finite-sample guarantee this paper certifies, so group-conditional remains the operative repair and RLCP is the higher-coverage option under a weaker guarantee. Venn–Abers matches the combiner at a stratum-conditional guarantee, so the repair does not require the gradient-boosted combiner.

	realized risk	coverage
Fixed ipTM ≥ 0.8 (field baseline)	0.301	0.69
Naive-transfer conformal	0.412	0.99
Accept-all	0.417	1.00
Group-conditional (combiner)	**0.190**	0.18
Localized / continuous novelty (RLCP) [5]	**0.187**	0.44
Per-stratum Venn–Abers [18]	**0.166**	0.24

realized risk 0.30 and naive-transfer conformal at 0.41 (Table 3). A randomized localized conformal gate (RLCP) reaches lower risk at higher coverage still (0.187 at 44%), but under a weaker, randomized marginal-over-localization guarantee rather than the stratum-conditional finite-sample one we require, so we keep group-conditional as the operative repair and report RLCP as the higher-coverage option a practitioner may prefer when a randomized guarantee suffices.

The label cost, as a design curve. Group-conditional repair spends target labels, and how many depends on what is being certified. For the binary pose label the count is a sufficient statistic and the exact binomial tail is its exact inversion, so across the 23 feasible cells it certifies at a median of 38 independent target-labels against 60 for a Hoeffding-Bentkus bound and 102 for pure Hoeffding. But that 38 is a *certification* sample size on already-feasible (familiar) strata, not the budget to push coverage one stratum outward. Figure 4 turns the latter into a design tool: certified coverage as a function of the label budget n_g, per stratum. With the native score, no budget buys usable coverage on the novel strata, because there the rule fails, not the estimate of it, the impossibility made empirical. The combined score, a partly novelty-aware re-score and so on the achievability side, unlocks the moderate strata: AF3 reaches 22% certified coverage on S_1 at $\sim$80 in-stratum labels and 83% with the full pool, and 77% on S_2, while S_3 needs close to the full label pool for any coverage at all. The combined score raises the ceiling, not the label floor. The budget is not just measured but predictable: for a stratum whose cleanest accept region has error rate $r < \alpha$, the exact binomial certifies non-zero coverage at (α, δ) once $n_g \geq \lceil \ln(1/\delta)/\mathrm{KL}(r\|\alpha) \rceil$ (binary relative entropy), a closed form a practitioner evaluates for their own α; it floors at 11 labels for a perfect accept region (r=0) and diverges as $r \uparrow \alpha$, which is exactly why

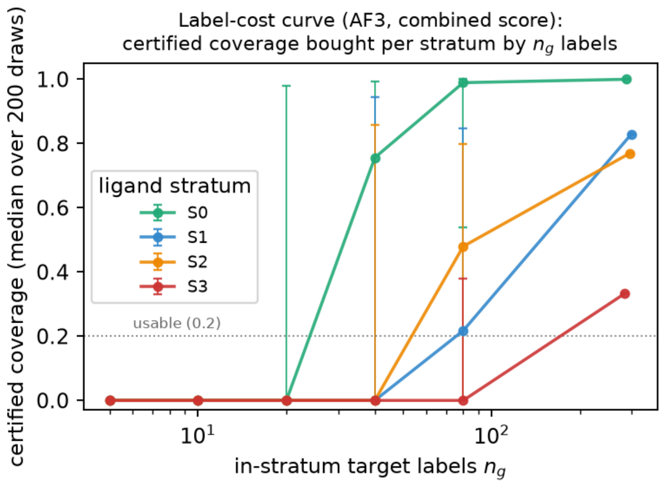

Figure 4: Label-cost curve. Certified coverage bought by n_g in-stratum target-labels, per ligand stratum (AF3, combined score, median over 200 draws with a 90% band). Moderate novelty (S_1, S_2) is certifiable at a measured label budget; the near-novel S_3 is label-starved even at the full pool.

a stratum whose best accept region already errs at $r \geq \alpha$ (every model's S_4; S_3 for Chai and Protenix) cannot be certified at any budget.

Deployment-computable stratifier. The repair needs a novelty stratum at deployment, but the oracle similarity is computed against the model's training set, which for a closed model like AF3 a user cannot enumerate. A public proxy built from the count of pre-cutoff PDB analogs (deposition recency and protein sequence identity are at chance) recovers the oracle stratification with rank correlation 0.59 and 83% adjacent agreement, and matches oracle per-stratum control on the low and moderate strata, but loosens control on the near-novel tail (true-S_3 risk $0.23 \to 0.37$; adjacent-stratum agreement over five bins is a lenient criterion, so this is the honest degradation). The binding limit, however, is not the proxy: the no-analog tail cannot be held even with the oracle stratifier, because its base error there is about 0.55–0.60, so a point-threshold gate keyed on the *true* stratum still realizes risk ≈ 0.60 and the certified Learn-then-Test group-conditional gate abstains outright (zero coverage) rather than issue a false certificate. So for closed-training models the repair is deployable one stratum out with a public proxy, and the extreme novel tail is uncertifiable by stratification alone, whether the stratifier is proxy or oracle. That, not a missing training set, is the honest ceiling.

A label-free worst-subpopulation certificate. When the novel subpopulation is unnamed we control the conditional value-at-risk of the accepted loss, the mean error of its worst $(1-\beta)$-fraction. For the binary label $\mathrm{CVaR}_\beta = \min(1, r/(1-\beta))$ with accepted error rate r, so a finite-sample bound on r [2] yields a CVaR_β bound at every β, and by CVaR–DRO duality this certifies that every accepted subpopulation of mass at least $m^\star = 1-\beta^\star$ has error $\leq \alpha$ [4], with no stratum labels and no importance weights, reparameterizing to a χ^2 robustness radius $\rho^\star$ [24]. A Bonferroni δ/K correction gives one certificate jointly across the five models (at 20% coverage, $m^\star \leq 0.52$ and $\rho^\star \geq 0.10$). This certificate is a real capability on moderate shift and, we report plainly, *vacuous exactly where it is most needed*: on deploy-to-novel $m^\star \to 1$ for most models, so the label-free certificate cannot rescue the novel tail either.

Match the certifier to the loss. On the *graded* loss $\min(\mathrm{RMSD}, 4)/4$ a variance-adaptive betting bound [30] certifies a usable operating point where a distribution-free bound certifies none: AF3 certifies 43% coverage at a 1 Å mean-RMSD target with realized capped-mean RMSD 0.90 Å, against essentially zero (0.04%) for a worst-case Hoeffding bound; the empirical guarantee coverage is 0.90 (46 of the 51 non-empty gates over 120 pose-level resamples met the target; at this resample count the estimate is consistent with the nominal $1-\delta = 0.90$ to within its Monte-Carlo error). The reliability layer is therefore not tied to the 2 Å convention, and "1 Å" is an actionable operating point where "$\alpha = 0.20$" is not.

Table 4: Certificate ledger. $K = 5$ models; joint claims split δ by a Bonferroni union bound.

Claim	δ	multiplicity	n (unit)	scope
Global / LOTO gate	0.10	fixed-sequence LTT	target-labels	per-model
Joint LOTO gate	0.02	Bonferroni δ/K	target-labels	joint (5 models)
CVaR / DRO bound	0.02	Bonferroni δ/K	accepted poses	joint (5 models)
Betting bound (graded)	0.10	anytime-valid (Ville)	accepted poses	per-model
Drift grid	0.10	Romano-Wolf step-down	55 cells	family-wise
Feasibility frontier (CP)	0.10	per-cell exact, uncorrected	40 cells	per-test
"For every model" effects	0.10	intersection-union	per-model p	no penalty

Certificate ledger. Table 4 states, per claim, the δ, the multiplicity method, the n, and whether the claim is joint or per-model, so no favorable δ is chosen per claim. The 55-cell drift grid (5 models $\times$ 11 axis-strata: 4 ligand + 4 pocket + 3 temporal non-reference strata) uses a Romano-Wolf step-down bootstrap [31] at family-wise 0.10; all 40 structural cells survive, an effect-dominated family, and only 2 of 15 temporal cells do.

A proposed deliverable: certificate cards. We propose packaging every repair verdict as a standardized card per (model, novelty stratum): α, δ, calibration n, accepted n, realized risk, exact Clopper-Pearson risk upper bound, retained coverage, reliability drift D, the recommended repair, and a three-way verdict (Fig. 5). We present the card as a proposal — its numbers are validated but its ergonomics are not user-tested, and the FEASIBLE / ABSTAIN-infeasible / ABSTAIN-underpowered distinction is exactly the kind a reader can collapse. The verdict distinguishes the two actions a practitioner must not confuse: a FEASIBLE card carries a certified operating point; an ABSTAIN-infeasible card means the rule provably fails (the Clopper-Pearson lower bound exceeds α at every coverage), so the target should be abandoned for this model; an ABSTAIN-underpowered card means the stratum is undecided for lack of labels, so the action is to collect them. The cards are the native-score view (the score a user has for free); the combined-feature group-conditional gate of Fig. 3 recovers usable coverage on some ABSTAIN-underpowered strata, which is why the two figures differ on, e.g., AF3 S_3 (native ABSTAIN, combined 5% certified).

7 Utility, priced and leakage-free

Ranking utility. Every utility number is target-grouped. Under a nested leave-one-target-out protocol (outer GroupKFold on system id; within each training fold a grouped fit/calibration split, so the combiner, the threshold, and the test target are mutually disjoint), the certificate is the per-fold Learn-then-Test guarantee, and we read it off the pooled Hoeffding-Bentkus upper bound rather than the fold count, which has little power (5 folds cannot resolve a 0.10 failure probability, so "4 of 5" is descriptive, not evidential). On that standard the combined-score gate keeps **73**% of AF3 poses at a pooled Hoeffding-Bentkus upper bound of $0.192 \leq \alpha$ (folds holding $4/5$, reported for context), whereas the native gate does *not* certify usable coverage leakage-free: at its 20% coverage only 1 of 2 non-empty folds holds and the pooled upper bound is $0.223 > \alpha$, and the Chai and Protenix native gates abstain outright (Table 5). So the honest leakage-free story is that the combined score is what makes the gate certifiable at usable coverage, not that it beats an equally-certified native gate; the earlier held-out pair (71% vs. 22%) is discarded because a pooled split leaks 95% of test targets into calibration where the grouped protocol shares zero. Grouping on system id still permits a homologous receptor in both folds, so we re-run the whole gate grouped on Runs N' Poses sequence cluster (954 clusters in the governed cohort): the AF3 headline is stable (0.72 combined coverage, still certified, HB upper bound 0.192), while two models lose coverage under the stricter grouping (Boltz-1 $0.60 \to 0.24$, Protenix $0.57 \to 0.41$), which we report rather than hide. At the stringent $\alpha = 0.10$ the combined gate certifies 13% of AF3 poses at realized risk 0.066 (pooled upper bound 0.094).

One map for every AF3 coverage number. Because the paper reports AF3 coverage under several protocols, Table 6 lists each with the experiment that produced it, so no two figures are mistaken for the same quantity. The 0.52-on-S_1 and 0.83-on-S_1 pair is the sharpest: the first is the coverage of a single

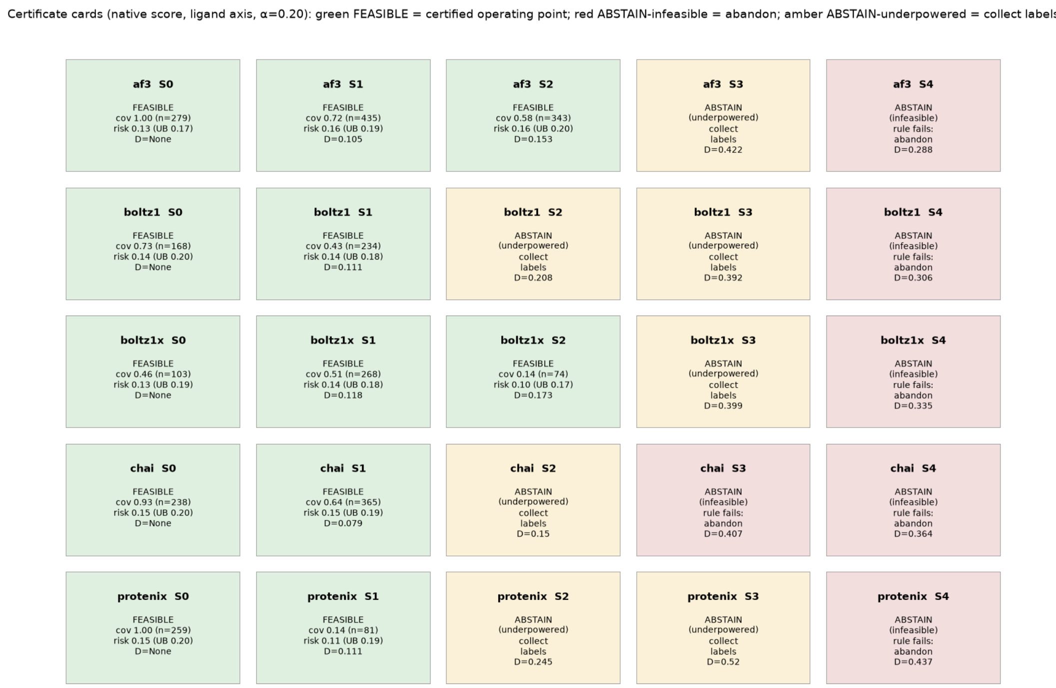

Figure 5: Certificate cards, native `ranking_score`, ligand axis, $\alpha = 0.20$, one per (model, stratum). Green FEASIBLE cards carry a certified operating point (retained coverage, accepted n, realized risk with its Clopper-Pearson upper bound, drift D); red ABSTAIN-infeasible cards mean the rule provably fails (abandon); amber ABSTAIN-underpowered cards mean undecided (collect labels). The combined-feature group-conditional gate (Fig. 3) is a different, higher-coverage protocol; the full grid across both novelty axes and both scores is in the SI.

α-certified per-stratum gate, the second is the maximum coverage the full in-stratum label pool certifies (the label-cost-curve endpoint) — different questions, same stratum.

A decision curve retires the choice of α. Rather than defend a level, we sweep the cost ratio $\lambda =$ cost(act on a wrong pose)/cost(abstain on a right pose) and plot net benefit (Fig. 6). The conformal gate is best for $\lambda \in [0.65, 4.55]$ (AF3; every model shows the same window), dominating a fixed-ipTM threshold at *every* λ and beating accept-all once wrong poses are more than $\sim 0.65\times$ as costly as a missed right one. Below $\lambda = 0.65$ accept-all wins (abstention only forfeits benefit when wrong poses are cheap), and above $\lambda = 4.55$ abstain-all wins (no accepted set is worth its false positives), so the gate occupies the intermediate, high-but-finite-cost regime, which is where a delivered pose that must be trusted lives.

A non-circular quality, honestly conditioned. Accepted-pose purity is $1-$ selective risk by construction, so it cannot show the gate buys anything past the guarantee. Interaction-fingerprint recovery is a different quantity a medicinal chemist reads: the fraction of the crystal's protein-ligand contacts a pose reproduces, where a contact is any receptor residue with a heavy atom within 4.5 Å of a ligand heavy atom, enumerated within each structure and matched to the crystal contact set by residue identity (no interaction typing or protonation modeling). Pooling all accepted against all rejected poses shows a large recall gap (0.19 for AF3), but most of that is the accepted set being enriched for low-RMSD poses. Conditioning removes the confound: *within* the sub-2 Å (correct) set, which holds 8,277 poses pooled across models, the untyped accepted-minus-rejected recall gap is $+0.02$ to $+0.04$ (all five 90% intervals excluding zero), stable on the single-chain subset. The chemically important question is whether this survives when contacts are *typed*: recomputed with a PLIP-style typed scheme (H-bond, salt bridge, π-stacking, hydrophobic; heavy-atom geometry, since the deposited coordinates carry no hydrogens), the within-correct gate lift is *largest on the polar, directional interactions* — H-bond $+0.072$ (AF3) and $+0.082$ (Chai), π-stacking $+0.073$ and $+0.061$, both intervals excluding zero and both exceeding the

Table 5: Leakage-free nested-LOTO gate, native vs. combined score ($\alpha = 0.20$, $\delta = 0.10$). The certificate is the per-fold Learn-then-Test guarantee; "folds" is the number of non-empty held-out folds whose realized risk landed $\leq \alpha$ (its validation), and $\mathrm{HB}^{\uparrow}$ is the more conservative pooled Hoeffding-Bentkus upper bound. The ✓ marks a pooled $\mathrm{HB}^{\uparrow} \leq \alpha$: it is a property of the aggregate bound alone, independent of the fold count, so a gate can hold in $4/5$ folds yet miss the ✓ (Boltz-1 native, HB 0.222) or hold in fewer folds and earn it (Boltz-1x native, $4/5$, HB 0.197). Chai/Protenix native gates abstain outright.

model	score	cov	risk	$\mathrm{HB}^{\uparrow}_{90}$	folds
AF3	native	0.20	0.189	0.223	$1/2$
	combined	**0.73**	0.176	0.192 ✓	$4/5$
Boltz-1	native	0.21	0.188	0.222	$4/5$
	combined	0.60	0.181	0.201	$4/5$
Boltz-1x	native	0.20	0.164	0.197 ✓	$4/5$
	combined	0.48	0.182	0.203	$3/4$
Chai-1	native			abstains	
	combined	0.64	0.178	0.196 ✓	$4/5$
Protenix	native			abstains	
	combined	0.57	0.182	0.201	$4/5$

Table 6: Every AF3 coverage and label figure in the paper, mapped to its protocol ($\alpha = 0.20$), so no two are read as the same quantity. The three "S_1/S_2" rows differ by protocol, not by contradiction: the frontier reach of an S_0-calibrated native rule is generous on the mildly-novel strata, per-stratum group-conditional recalibration is stricter, and the label-cost maximum is what the full in-stratum label pool certifies.

Quantity	Protocol	Scope	coverage	risk
Combined gate, marginal	nested target-grouped LOTO	all strata	0.73	0.176
Native gate, marginal	nested target-grouped LOTO	all strata	0.20	0.189
Frontier / certificate card (native)	S_0-calibrated rule, max reach	S_1 / S_2	0.72 / 0.58	$\leq \alpha$
Group-conditional (native)	per-stratum α-certified gate	S_1 / S_2	0.08 / 0.12	$\leq \alpha$
Group-conditional (combined)	per-stratum α-certified gate	S_1 / S_2	0.52 / 0.39	$\leq \alpha$
Deploy-to-novel (combined)	calibrate familiar, deploy novel	S_2, S_3	0.18	0.190
Label-cost maximum (combined)	full in-stratum label pool	S_1 / S_2	0.83 / 0.77	$\leq \alpha$
Label budgets (not coverages): certification sample size = median 38 over the 23 feasible cells; one-stratum-out extension budget $\approx$ 80 in-stratum labels (the label-cost curve).				

hydrophobic gap (+0.05, +0.04). So the lift is not merely hydrophobic burial: among equally-correct poses, the gate-accepted ones recover the SAR-relevant contacts a medicinal chemist reads markedly better than the rejected ones. Separately, the accepted set is PoseBusters-valid at a higher rate than the rejected set for every model (AF3 0.77 vs. 0.58), a free strengthening; certifying the joint label (RMSD $\leq$ 2 Å *and* physically valid) directly is expensive because the confidence ranks for correctness, not validity, and we report that cost rather than hide it.

8 Generality and limitations

Pseudo-prospective. Freezing calibration on everything deposited before 2023-05-31 and deploying on everything after (per model $\sim$1300 pre- and $\sim$900 post-cutoff targets), with no stratum information at calibration: on the native gate, AF3, Boltz-1x, and Protenix hold the post-cutoff certificate (realized-risk upper bound $\leq \alpha$), Boltz-1 overshoots by its upper bound (0.207), and Chai abstains; on the combined gate, AF3 and Boltz-1 hold, Chai overshoots (0.202), and Boltz-1x and Protenix abstain. Abstention is a non-certification, not a violation, and every held gate's realized post-cutoff risk stays within 0.01 of its pre-cutoff expectation (the realized-minus-expected gap ranges -0.04 to $+0.003$ across models), so temporal deployment does not silently break control. This is a recency transfer, complementary to the structural story, and inherits the caveat that Runs N' Poses is entirely post-cutoff.

A newer-cutoff model, and seed stability. Run as a fully governed model with its own 2023 similarity

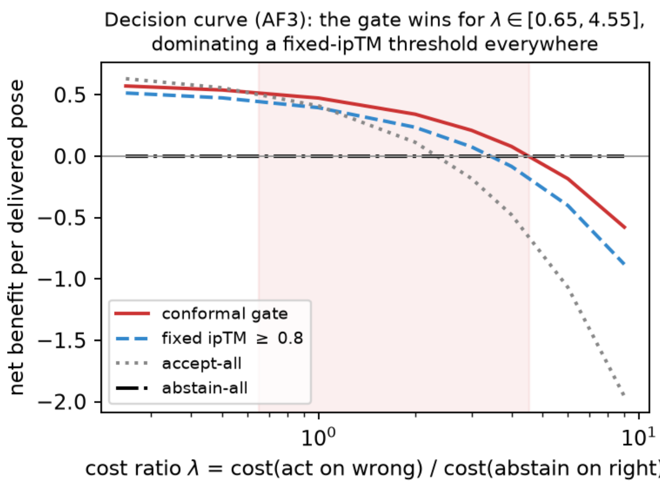

Figure 6: Decision curve (AF3). Net benefit per delivered pose vs. the cost ratio λ; the shaded band is where the conformal gate is the best of the four strategies. It dominates a fixed-ipTM threshold everywhere and serves the high-cost regime, so there is no single α to defend.

reference, Boltz-2 (a 2023-cutoff model) shows the same break: its marginal risk of 0.17 hides worst-stratum realized risk 0.41 (ligand) and 0.51 (pocket) against $\alpha = 0.20$, with S_3 drift $+0.40$ (ligand) and $+0.59$ (pocket, 2023-referenced), CIs excluding zero. The collapse is therefore not an artifact of the 2021 cohort. The gate is also deployment-stable across the diffusion randomness that produces the pose: refitting the AF3 certified gate within each of the five diffusion seeds moves the threshold by 0.002 and the coverage by 0.02, and gating a random sample per target rather than the top-1 changes coverage by ~ 0 at the same certified risk, so the certificate is not an artifact of the delivery policy.

Cross-dataset. Recast as risk control rather than ranking, a threshold frozen on Runs N' Poses and deployed on regenerated FoldBench [9] Protenix predictions accepts only 3 of the 52 low-homology targets (coverage 6%, Clopper-Pearson risk upper bound 0.63), so the low-homology arm is uninformative about risk and informative only about coverage *starvation*, the coverage collapse of Sec. 3 reproduced on a second dataset; on the 384 train-similar targets the frozen gate accepts 18 at realized risk 0.11. The interface-ipTM gate out-ranks a `ranking_score` control (AURC 0.380 vs. 0.454) but their 90% bootstrap intervals overlap, and the regenerated model's self-scored top-1 success (0.40) differs from the released 0.57, so this stays a directional transfer check, feature and label self-consistent within the one run. A third dataset would need a benchmark that releases ipTM and PoseBusters per pose; where none is available we state that concrete community ask rather than overclaim generality.

Label quality is not the confound. The label is a 2 Å BiSyRMSD against a single deposited crystal pose, inheriting the field's pose-correctness convention [32]. We tested whether crystallographic quality drives the break: fetching resolution for all 2,412 entries, the median resolution is flat across strata (S_0 1.98, S_3 2.00 Å, novel targets are not lower-resolution), the S_3 break persists on high-resolution-only entries ($0.42 \to 0.46$ if anything), and it survives 5% and 10% random label flips (S_3 risk stays above α in 100% and 99% of resamples). So the novel-pocket collapse is a property of confidence reliability, not label noise; the graded-loss betting bound (Sec. 6) additionally removes the hard 2 Å cutoff.

Limitations. The no-computable-analog stratum S_4 is a mixture (Sec. 2): ~ 29 of 76 instances are a genuine drug-like no-analog core and the rest are a similarity-search coverage gap, so its "uncertifiable" reading applies cleanly only to that core, and we lead every claim on S_3. The best ranker in our screening analysis (SI), a Boltz-2 affinity head, carries no coverage guarantee: extending the framework to certify an affinity ranker (which needs an affinity-error label) is the top open problem. All evaluation is retrospective; a prospective screen with wet-lab confirmation is the natural next step.

What to do with this, concretely. The layer is not only an abstention rule; the label-cost curve is a plan. A worked example: a team running AlphaFold3 on a congeneric series whose scaffold is one novelty stratum out from the training set (an S_1–S_2 regime) should not trust the raw ranking score — it certifies almost nothing there leakage-free — but can recover a certified $\sim 50\%$ acceptance rate by building a group-conditional gate, which costs on the order of 80 solved structures of in-series analogs to calibrate. If the series is genuinely novel (S_3–S_4, no close training analog), the certificate card reads ABSTAIN-infeasible and the honest recommendation is to treat every delivered pose as unverified and fall back to experiment. The deliverable is the card that tells the team which of these three situations it is in, before it commits medicinal-chemistry effort to a wrong pose.

9 Conclusion

Co-folding confidence becomes a risk-controlled decision only when the guarantee is audited where it fails. On structural novelty a marginally valid gate that calibrates on familiar targets and deploys on novel ones realizes error 0.55 against a 0.20 target (about 2.7×; the frozen score still *ranks* poses there, AUROC 0.71, but the base rate is too low for any threshold to certify), and 13 of 40 model-by-stratum cells admit no threshold on any score in the family — global, interface, or ligand-local — that is certifiable at any coverage. An exact decomposition explains why: the score-fiber gap that opens there closes under in-stratum recalibration but under no training-free reweighting. Three honest bottom lines follow. Co-folding confidence is certifiable on the familiar strata (where the exact binomial certifies from a median of 38 labels over the 23 feasible cells, against 102 for Hoeffding). It is extendable one stratum outward at a measured price, on the order of 80 in-stratum target structures for usable certified coverage, and with a public proxy stratifier for closed-training models. And on the extreme novel tail, exactly where a program most needs a guarantee, neither stratification nor a label-free certificate can supply one, so abstention is the certified action. The layer runs on frozen models' released predictions on CPU and ships as certificate cards per model and novelty stratum.

Data and code availability. Code and calibration tables are at `https://github.com/rikhinkavuru/foldgate` under an OSI-approved license; a one-command `make repro` reproduces the tabular results from a pinned environment (`uv.lock`, fixed seeds). We ship a single per-pose analysis table (`analysis_table.csv`, one row per delivered pose with the scores, the label, both similarities and strata, sequence cluster, crystal resolution, and ligand-local pLDDT) from which every tabular result in the paper can be re-derived without the structure stream. Runs N' Poses is at Zenodo (`10.5281/zenodo.14794785`); the regenerated FoldBench predictions and self-scored labels are at Zenodo (`10.5281/zenodo.21417241`); the screening scores are from **(author?)** [10]. The coverage grid, fixed-sequence order, binning, and endpoints are frozen in a pre-registration file, timestamped independently of the git history (OpenTimestamps), in the repository.

Funding. No external funding. **Competing interests.** The author declares none. **Broader impact.** The layer is a decision gate for drug-discovery pipelines; its central message is conservative (abstain where the guarantee is vacuous), and its main risk is misuse as a certificate on the novel tail, which Secs. 4–6 explicitly forbid.

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A Proof of the decomposition, and the ties bracket

Setup. Frozen score $s = s(X)$, error label $Y \in \{0,1\}$, accept region $A_\tau = \{s \geq \tau\} \in \sigma(s)$. We work with the score-conditional error rates $\eta_P(s) = \mathbb{E}_P[Y \mid s]$ and $\eta_Q(s) = \mathbb{E}_Q[Y \mid s]$, since a threshold gate and a density-ratio weight on the score both live in $\sigma(s)$; a training-free reweighting has weights $w = w(s) \geq 0$ that are $\sigma(s)$-measurable and use no target labels. Assume s has no atom at the operating threshold, so for each coverage c there is a unique τ_c with $Q(s \geq \tau_c) = c$ (on the top-k rank lattice this is τ_c up to the $1/n$ granularity); real scores carry ties handled by boundary randomization, under which the equalities become $\leq$ / $\geq$ brackets, and Sec. 5 bounds the bracket empirically (0 on the ranking-score gate, $\leq 0.7\%$ coverage on ipTM).

Lemma (a score-measurable reweighting sees only the source score-conditional). For any such w, the reweighted plug-in selective risk is $\widetilde{E}^w(\tau) = \mathbb{E}_P[wY\mathbf{1}_{A_\tau}]/\mathbb{E}_P[w\mathbf{1}_{A_\tau}] = \mathbb{E}_{Q_w}[\eta_P(s) \mid A_\tau]$, with $dQ_w \propto w\,dP_s$. *Proof.* w and $\mathbf{1}_{A_\tau}$ are $\sigma(s)$-measurable, so tower over s: $\mathbb{E}_P[wY\mathbf{1}_{A_\tau}] = \mathbb{E}_P[w\,\eta_P(s)\,\mathbf{1}_{A_\tau}]$; normalize. The target score-conditional $\eta_Q(s)$ cannot appear. □

Proposition. At coverage c: (a) $R_Q(\tau_c) = R_{\text{ref}}(\tau_c) + \bar{\Delta}_c$ with $R_{\text{ref}}(\tau_c) = \mathbb{E}_Q[\eta_P(s) \mid A_{\tau_c}]$ and $\bar{\Delta}_c = \mathbb{E}_Q[\eta_Q(s) - \eta_P(s) \mid A_{\tau_c}]$; (b) *coverage pinning:* any rule — any weight, $\sigma(s)$-measurable or not — that realizes coverage c on Q accepts exactly A_{τ_c}, so its realized risk equals $R_Q(\tau_c)$, independent of the weight; (c) with the oracle score-weight $w^\star = dQ_s/dP_s$ the Lemma's certificate equals $R_{\text{ref}}(\tau_c)$, under-reporting the realized risk by exactly $\bar{\Delta}_c$. *Proof.* (a) add and subtract $\mathbb{E}_Q[\eta_P(s) \mid A_{\tau_c}]$. (b) the no-atom assumption pins τ_c from c, so realized risk is a functional of (Q, τ_c) alone. (c) Lemma with $w^\star$ gives $Q_{w^\star} = Q$. $\square$ (A part-(d) "α achievable at coverage c iff $\bar{\Delta}_c \le \alpha - R_{\text{ref}}$" is only (a)+(b) rearranged and we do not count it as a result; achievability by *lowering* coverage is the empirical frontier of Sec. 4.) The statement binds only scalar thresholds of a fixed score with score-measurable weights; re-scoring on ν or a per-stratum threshold escapes it (the achievability side), and in-stratum split calibration [23] gives stratum-conditional miscoverage $\le 1/(n_g{+}1)$ regardless of drift; conformal risk control [26] gives the same slack for the joint accept-and-err rate, and the ratio functional (a random-denominator selective risk) is controlled by Learn-then-Test at the finite-sample cost the label-cost curve measures.

B Benchmark and validation details

The synthetic generator draws a stratum from a tilted ν-marginal, a score logit $z \sim \mathcal{N}(m_0 - E\nu, \sigma^2)$, and a label with $P(Y{=}1) = \varepsilon + (1-2\varepsilon)\,\text{sigmoid}(\beta_0 + \beta_1 z - D_{\text{eff}}\nu)$, isolating covariate shift (E), concept drift (D), and an error floor (ε); every proposition quantity is available in closed form and cross-checked by Monte Carlo. The real benchmarks reduce any classifier to (s, y, ν); realized worst-stratum selective risk at $\alpha = 0.20$: elec2 marginal 0.31, weighted-conformal 0.34, group-conditional 0.20; ACS Income marginal 0.23, weighted-conformal 0.20, group-conditional 0.20, the reliability drift separating the two being 0.36 (elec2, concept) against 0.05 (ACS, covariate).

C A benchmark-integrity note for cross-model pose geometry

Recomputing pose RMSD in a common frame carries a silent trap. Runs N' Poses ships only the system's receptor chain while co-folding models predict the full biological assembly, so a chain-ordinal superposition can land the ligand in the wrong protomer of a homodimer: the backbone still fits (superposition residual ~ 0.3 Å) while the ligand sits tens of Ångström away. Neither the residual nor a triangle-inequality consistency check detects it, and a distance-consistency check passed at $0/13{,}146$ while 21% of recomputed distances and 15% of binary correctness labels were wrong. Only comparison against the shipped BiSyRMSD label exposes it; restricting to single-chain receptors with an unambiguous ligand copy raises agreement from Spearman 0.620 to 0.995 and defines the 6,163-instance subset any recomputed-geometry result in this paper uses (Table 2). We flag this for reuse; a symmetry-aware chain assignment is the general fix.